

Emre Arslan

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EDUCATION

Brown University, *B.S. Computer Science*, 3.9/4.0 GPA

Providence, RI | **Expected Graduation May 2025**

Relevant Courses: Data Structures & Algorithms, Deep Learning, 3D Photography, Computer Graphics, Parallel Computing

SAT Overall: 1550

Uskudar American Academy High School, 4.00/4.00 GPA

Istanbul, Turkey | Class of 2021

EXPERIENCE

Brown University, *Computer Vision Research Assistant* ([link](#))

September 2024 - Present

- Researched novel optimizations to accelerate 3D Gaussian Splatting training and profiled GPU costs with NVIDIA Nsight.
- Implemented dilation rendering and increased thread utilization in CUDA to speed up total training by 22%.
- Contributed as 4th author to a research paper to Conference on Computer Vision and Pattern Recognition (CVPR).

Apple, *Vision Pro Software Engineering Intern - Persona*

June 2024 - August 2024

- Reduced the runtime and power usage of GPU kernels for 3D spatial Personas on the Apple Vision Pro.
- Developed Metal shaders, cutting GPU megacycles by 64% and resulting in significant real-time rendering efficiency gains.
- Researched and crafted optimization algorithms using PyTorch, C++, and custom MPS kernels.

Brown University, *Computer Systems Research Assistant*

June 2023 - April 2024

- Engineered and presented demo applications on streaming/vector databases and Retrieval-Augmented Generation (RAG).
- Designed retrieval pipelines for text similarity search and anomaly detection using Faiss, Milvus and Hugging Face Transformers, increasing query performance by 30%.

PERSONAL PROJECTS

Real-Time Raytraced Website ([link](#))

2024

- Developed a real-time raytraced 3D landing page with interactive graphics using WebGPU compute shaders.

Gfx-Main ([link](#))

2023

- Designed a high-level real-time graphics library with WebGPU API and TypeScript, enabling developers to easily create 3D geometries and handle complex rendering tasks.
- Divided the API to let users create 5+ shape meshes or upload .obj files, and create lights, textures, materials and cameras.
- Added support for 3 lighting models with WGSL shaders as well as depth buffering and multisample anti-aliasing.

ACADEMIC PROJECTS

Monte Carlo Path Tracer

2025

- Coded a physically-based renderer in C++ using the path tracing algorithm to simulate realistic global illumination.
- Developed advanced rendering features including soft shadows, color bleeding, caustics, and image-based lighting.

Stable NeRF (Group Lead of 4, [link](#))

2024

- Integrated Neural Radiance Fields (NeRF) with Stable Diffusion to create geometrically consistent novel view generation.
- Utilized the Instant NGP architecture to formulate a 4-channel latent NeRF for deep 3D features with Tiny-CUDA-NN.
- Trained IP Adapters and a CNN to fine-tune and condition Stable Diffusion, enabling it to leverage 3D inputs for generation.
- Validated the framework through quantitative and qualitative analyses on the Objaverse and NeRF synthetic datasets.

Surface-Aligned Gaussian Splats (Group Lead of 2)

2024

- Implemented mesh surface aligned Gaussian splatting in PyTorch with the gsplat library's renderer, demonstrating 15% improved performance over point cloud-based methods.
- Utilized COLMAP to calibrate cameras, estimate point clouds, and generate base template meshes of objects from images.
- UV unwrapped the template mesh and assigned Gaussian splats to texels, replacing point cloud-based initialization.

3D Poisson Equation Solver ([link](#))

2023

- Built a 3D Poisson equation solver in C++ utilizing MPI, OpenMP, and CUDA to achieve high parallelism, achieving significant performance gains over single-node or CPU-only systems.
- Optimized halo exchange between distributed nodes using CUDA-aware MPI, enhancing data transfer efficiency and reducing communication overhead.
- Demonstrated that the Red-Black solver can improve computation and converge up to 110% faster than the Jacobi solver.

SKILLS & INTERESTS

Programming Languages: Python, JavaScript/TypeScript, C/C++

Frameworks/Tools: PyTorch, TensorFlow, NumPy, CUDA, WebGPU, Metal, OpenGL, Eigen, LangChain, Hugging Face

Language: Bilingual proficiency in Turkish and English

Interests: Brown University Table Tennis team member, guitar player and vocalist

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ADDITIONAL PROJECTS

Mesh Processing Engine

2025

- Built a mesh processing engine in C++ supporting operations such as subdivision, simplification, remeshing, and denoising.
- Designed and implemented a half-edge data structure to efficiently manage and manipulate mesh topology.

3D Digital Laser Scanner

2024

- Engineered software in C++ to process images from a 3D digital laser scanner and generate high-accuracy digital scans.
- Calibrated the camera, turntable, and laser plane to ensure precise alignment and measurement.
- Designed laser line detection algorithms and performed triangulation to reconstruct 3D points from scanned data.

Neural Radiance Fields (group lead of 4, [link](#))

2024

- Re-implemented the Neural Radiance Fields model for 3D novel view synthesis in PyTorch, achieving high-quality on the original NeRF dataset objects and custom objects made by the team.
- Trained a multilayer perceptron, volume-rendered complex 3D scenes, and presented design choices and performance evaluations to an audience of 300+ deep learning students.

3D Gaussian Splatting Camera Optimization (group lead of 4)

2023

- Leveraged 3D Gaussian Splatting for Real-Time Radiance Field Rendering to perform AI-driven camera optimization.
- Integrated the ResMem network from computational neuroscience to evaluate the memorability of rendered images.
- Optimized camera positioning based on image memorability scores to efficiently search the scene for highest visual impact, using both a PyTorch network and Monte Carlo sampling.

Realtime Renderer

2022

- Architected a real-time renderer in C++ using OpenGL, featuring geometry mesh generation, Phong lighting, and an interactive virtual camera for dynamic scene exploration.